

## SILIFKE, Turkey

WMO: 173300

Lat: 36.3821N

Lon: 33.9366E

Elev: 49

StdP: 14.67

Time Zone: 3.00 (E03)

Period: 94-17

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 38.0       | 40.4 | 10.1                          | 9.2 | 45.3 | 14.9 | 11.7 | 45.8 | 18.0                  | 40.0 | 16.5 | 41.7 | 9.6                   | 320  | 0.329 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 14.6                   | 94.9            | 74.8 | 93.0 | 76.7 | 91.5 | 77.0 | 82.3                | 90.5 | 81.2 | 89.5 | 80.2 | 88.5 | 5.0                  | 140  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 80.0                            | 155.9 | 88.5 | 78.9 | 150.1 | 87.4 | 77.7 | 144.2 | 86.6 | 46.3          | 90.5 | 45.1 | 90.1 | 43.9 | 88.4 | 86.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |     |                    | n-Year Return Period Values of Extreme Temperature |           |     |            |     |            |     |            |     |
|-------------------|------|------|----------------------------|------|-----|--------------------|----------------------------------------------------|-----------|-----|------------|-----|------------|-----|------------|-----|
| 1%                |      | 2.5% | 5%                         | Mean |     | Standard Deviation |                                                    | n=5 years |     | n=10 years |     | n=20 years |     | n=50 years |     |
| Min               |      | Max  |                            | Min  | Max |                    | Min                                                | Max       | Min | Max        | Min | Max        | Min | Max        |     |
| (a)               | (b)  | (c)  |                            | (e)  | (f) | (g)                | (h)                                                | (i)       | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |
| 14.3              | 12.2 | 10.7 |                            | N/A  | N/A | N/A                | N/A                                                | N/A       | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |
|                   |      |      | DB                         | N/A  | N/A | N/A                | N/A                                                | N/A       | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |
|                   |      |      | WB                         | N/A  | N/A | N/A                | N/A                                                | N/A       | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 68.2   | 52.0  | 53.2  | 58.6  | 64.1  | 71.9  | 78.5  | 83.4  | 83.8  | 79.4  | 74.2  | 63.3  | 54.7  | (6)  |
|                                                                        | DBStd       | 12.30  | 4.95  | 5.45  | 5.13  | 4.81  | 4.72  | 3.98  | 2.60  | 2.76  | 3.13  | 4.90  | 6.09  | 5.34  | (7)  |
|                                                                        | HDD50       | 97     | 40    | 28    | 4     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 22    | (8)  |
|                                                                        | HDD65       | 1439   | 404   | 329   | 208   | 72    | 2     | 0     | 0     | 0     | 0     | 5     | 98    | 321   | (9)  |
|                                                                        | CDD50       | 6733   | 100   | 119   | 270   | 423   | 679   | 854   | 1036  | 1049  | 883   | 751   | 403   | 166   | (10) |
|                                                                        | CDD65       | 2601   | 0     | 0     | 9     | 45    | 216   | 404   | 571   | 584   | 433   | 291   | 48    | 0     | (11) |
|                                                                        | CDH74       | 27396  | 0     | 0     | 42    | 291   | 1627  | 4183  | 7447  | 7577  | 4151  | 1943  | 135   | 0     | (12) |
|                                                                        | CDH80       | 11519  | 0     | 0     | 3     | 70    | 458   | 1574  | 3536  | 3668  | 1622  | 578   | 10    | 0     | (13) |
| (14) Wind                                                              | WSAvg       | 3.9    | 5.0   | 5.1   | 3.9   | 3.7   | 3.2   | 3.4   | 3.3   | 3.1   | 3.7   | 3.8   | 3.9   | 4.2   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 20.4   | 3.7   | 2.8   | 1.7   | 1.1   | 1.1   | 0.2   | 0.1   | 0.1   | 0.4   | 1.4   | 3.0   | 4.9   | (15) |
|                                                                        | PrecMax     | 37.2   | 9.8   | 4.7   | 4.3   | 3.0   | 4.4   | 1.0   | 0.9   | 1.1   | 2.4   | 3.7   | 8.3   | 19.1  | (16) |
|                                                                        | PrecMin     | 11.7   | 0.0   | 0.1   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | (17) |
|                                                                        | PrecStd     | 6.8    | 2.4   | 1.1   | 1.2   | 0.8   | 1.2   | 0.3   | 0.2   | 0.2   | 0.6   | 1.1   | 2.4   | 4.2   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 66.5  | 69.7  | 78.5  | 86.7  | 93.7  | 98.7  | 98.9  | 97.4  | 95.4  | 92.7  | 81.2  | 70.0  | (19) |
|                                                                        |             | MCWB   | 54.9  | 56.4  | 57.3  | 65.9  | 72.3  | 74.8  | 73.4  | 76.6  | 73.9  | 64.5  | 64.3  | 56.6  | (20) |
|                                                                        | 2%          | DB     | 63.7  | 66.9  | 73.9  | 81.2  | 88.6  | 92.8  | 94.5  | 94.1  | 91.3  | 88.4  | 77.7  | 67.2  | (21) |
|                                                                        |             | MCWB   | 54.1  | 55.3  | 56.6  | 65.0  | 69.9  | 72.0  | 75.6  | 78.8  | 73.7  | 67.1  | 63.1  | 55.6  | (22) |
|                                                                        | 5%          | DB     | 61.5  | 64.3  | 70.6  | 77.1  | 84.9  | 89.4  | 92.6  | 92.7  | 89.2  | 85.6  | 74.9  | 64.4  | (23) |
|                                                                        |             | MCWB   | 52.8  | 53.9  | 56.0  | 62.9  | 69.7  | 72.6  | 76.6  | 79.6  | 74.6  | 67.7  | 61.7  | 54.2  | (24) |
|                                                                        | 10%         | DB     | 59.2  | 61.9  | 67.7  | 73.4  | 81.9  | 86.9  | 90.9  | 91.3  | 87.5  | 83.1  | 71.9  | 61.7  | (25) |
|                                                                        |             | MCWB   | 51.0  | 52.4  | 55.5  | 62.1  | 68.4  | 73.1  | 77.2  | 79.7  | 73.9  | 66.8  | 60.3  | 52.7  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 57.9  | 58.5  | 62.1  | 68.5  | 77.0  | 79.9  | 83.0  | 84.2  | 81.9  | 75.8  | 68.0  | 60.0  | (27) |
|                                                                        |             | MCDB   | 63.2  | 66.4  | 71.0  | 79.8  | 87.8  | 89.3  | 91.6  | 91.7  | 90.7  | 84.7  | 77.1  | 65.9  | (28) |
|                                                                        | 2%          | WB     | 55.9  | 56.8  | 59.5  | 66.3  | 74.0  | 78.2  | 81.8  | 82.8  | 79.4  | 73.7  | 65.4  | 57.9  | (29) |
|                                                                        |             | MCDB   | 61.7  | 64.6  | 68.3  | 76.6  | 83.2  | 86.8  | 90.3  | 90.6  | 88.1  | 83.5  | 74.5  | 63.7  | (30) |
|                                                                        | 5%          | WB     | 54.2  | 55.2  | 58.1  | 64.7  | 71.6  | 77.1  | 80.8  | 81.8  | 77.9  | 71.6  | 63.6  | 56.2  | (31) |
|                                                                        |             | MCDB   | 59.4  | 62.7  | 67.3  | 74.9  | 80.4  | 85.2  | 89.1  | 89.8  | 86.1  | 81.5  | 72.1  | 61.4  | (32) |
|                                                                        | 10%         | WB     | 52.7  | 53.4  | 56.6  | 63.1  | 70.0  | 76.0  | 79.7  | 80.8  | 76.4  | 69.3  | 61.9  | 54.6  | (33) |
|                                                                        |             | MCDB   | 57.5  | 60.2  | 66.2  | 71.9  | 79.4  | 83.8  | 87.9  | 89.1  | 84.6  | 79.5  | 69.6  | 59.7  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 11.7  | 13.0  | 14.7  | 15.8  | 16.4  | 15.5  | 14.8  | 14.6  | 15.9  | 16.3  | 14.3  | 11.8  | (35) |
|                                                                        |             | MCDBR  | 14.7  | 16.1  | 17.9  | 20.1  | 20.0  | 19.0  | 17.1  | 15.7  | 17.4  | 18.7  | 16.9  | 15.0  | (36) |
|                                                                        |             | MCWBR  | 9.2   | 10.0  | 9.9   | 10.7  | 11.3  | 11.7  | 11.4  | 10.8  | 12.2  | 11.2  | 10.4  | 9.1   | (37) |
|                                                                        | 5% WB       | MCDBR  | 12.8  | 15.3  | 16.3  | 17.5  | 18.0  | 15.4  | 15.3  | 15.4  | 15.9  | 17.6  | 16.1  | 13.1  | (38) |
|                                                                        |             | MCWBR  | 8.6   | 10.1  | 10.4  | 10.6  | 11.7  | 10.0  | 9.9   | 10.5  | 11.7  | 12.3  | 10.7  | 8.7   | (39) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.371 | 0.396 | 0.434 | 0.489 | 0.473 | 0.434 | 0.448 | 0.456 | 0.418 | 0.417 | 0.385 | 0.369 | (40) |
|                                                                        | taud        |        | 2.315 | 2.239 | 2.132 | 2.015 | 2.118 | 2.276 | 2.249 | 2.214 | 2.318 | 2.275 | 2.314 | 2.341 | (41) |
|                                                                        | Ebn at Noon |        | 255   | 263   | 263   | 254   | 260   | 270   | 265   | 260   | 265   | 252   | 247   | 247   | (42) |
|                                                                        | Edh at Noon |        | 31    | 38    | 46    | 55    | 50    | 43    | 43    | 44    | 38    | 36    | 31    | 29    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 731   | 1016  | 1467  | 1840  | 2163  | 2475  | 2483  | 2226  | 1835  | 1323  | 941   | 696   | (44) |
|                                                                        | RadStd      |        | 88    | 107   | 85    | 130   | 102   | 96    | 58    | 50    | 71    | 85    | 68    | 87    | (45) |

## Historical Trends

|      |                         | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (16 neighbors) | +0.52 | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | +184  | N/A   |

Nomenclature: See separate page